Chapter 3 Convergence of fourier series

2. prove $l^2(\mathbb{Z})$ is complete

pf: let $A_k = \{a_{km}\}_{n\in\mathbb{Z}}$ $k=1,2,\dots$ be a Cauchy sequence

that is $\forall \varepsilon>0$ $\exists N$ $\forall k>l>N$ we have $\|A_k - A_l\| < \varepsilon$

* thus $\sum_{n\in\mathbb{Z}} |a_{k,n} - a_{l,n}| < \varepsilon$ In particular: $|a_{k,n} - a_{l,n}| < \varepsilon$

$\Rightarrow \{a_{k,n}\}$ $k=1,2,\dots$ has a limit b_n

Consider $B = (\dots, b_{-1}, b_0, b_1, \dots)$

$\|A_k - A_l\| < \varepsilon$. let $k\to\infty$ we have $\|A_l - B\| < \varepsilon \Rightarrow \lim_{l\to\infty} A_l = B$

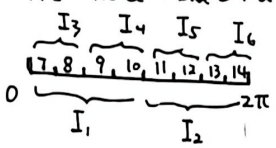
$\|B\| \le \|A_l\| + \|A_l - B\| < \infty$. $\Rightarrow B \in l^2(\mathbb{Z})$

So $l^2(\mathbb{Z})$ is complete ☒.

3. Construct a sequence of integrable functions $\{f_k\}$ on $[0,2\pi]$ s.t.

$\lim_{k\to\infty} \frac{1}{2\pi}\int_0^{2\pi} |f_k(\theta)|^2 d\theta = 0$. but $\lim_{k\to\infty} f_k(\theta)$ fails to exist $\forall \theta\in[0,2\pi]$.

pf: ~~We notice with rational numbers~~



choose I_k like this. taking $f_k = \chi_{I_k}$

then $\lim_{k\to\infty} \frac{1}{2\pi}\int_0^{2\pi} |\chi_{I_k}(\theta)|^2 d\theta = 0$.

but $\lim_{k\to\infty} f_k(\theta)$ fails to exist $\forall\, \theta\in[0,2\pi]$

4. the vector space $\mathcal{R}$ of integrable functions with inner product

$(f,g) = \frac{1}{2\pi}\int_0^{2\pi} f(x)\overline{g(x)}\,dx$

(a). $\exists f\in\mathcal{R}$ s.t. $f\not\equiv 0$. $\|f\| = 0$.

(b). If $\|f\|_1 = 0$. then $f(x) = 0$ whenever f is continuous at x.

(c). If $f\in R$. vanishes at all of its points of continuity then $\|f\| = 0$.

(a), (b) is trival.

(c) Since f is integrable. $\therefore$ f has at most countable discontinuities. Ω.

then cover them with $\mathcal{O}$ which length is 0, then we will have $\|f\| = \frac{1}{2\pi}\left(\int_0^{2\pi} - \int_\Omega\right)|f(x)|\,dx$

$= 0$ ☒

5. Let $f(\theta) = \begin{cases} 0 & \theta = 0 \\ \log(\frac{1}{\theta}) & 0<\theta\le 2\pi. \end{cases}$

define a sequence $f_n(\theta) = \begin{cases} 0 & \text{for } 0\le\theta\le\frac{1}{n} \\ f(\theta). & \text{for } \frac{1}{n}<\theta\le 2\pi \end{cases}$

prove that $\{f_n\}_{n\ge 1}$ is a Cauchy Sequence in $\mathcal{R}$ However, $f\notin\mathcal{R}$.

18 pf: $\|f_n - f_m\|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f_n - f_m|^2 \, dx$ $\quad n>m$



$= \frac{1}{2\pi}\int_{\frac{1}{n}}^{\frac{1}{m}} [\log(\frac{1}{\theta})]^2 d\theta$ + [illegible]

$= -\frac{1}{2\pi}\int_{\frac{1}{n}}^{\frac{1}{m}} (\log\theta)^2 d\theta = -\frac{1}{2\pi}\,\theta(\log\theta)^2 - 2\theta\log\theta + 2\theta \Big|_{\frac{1}{n}}^{\frac{1}{m}} \to 0 \quad m\to\infty$

$\int (\log\theta)^2 d\theta = \theta(\log\theta)^2 - 2\int \log\theta\, d\theta = \theta(\log\theta)^2 - 2\theta\log\theta + 2\theta$

So $\{f_n\}$ is a Cauchy sequence However f is not bounded

Such that $\mathcal{R}$ is not complete ☒

16 Consider $\{a_k\}_{k\in\mathbb{Z}}$ defined by: $a_k = \begin{cases} \frac{1}{k} & \text{if } k\geq 1 \\ 0 & \text{if } k\leq 0 \end{cases}$

Note that $\{a_k\} \in l^2(\mathbb{Z})$, but that no Riemann integrable function has k^{th} fourier coefficient equal to a_k for all k.

pf: $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx} dx = a_n$

Then by Abel means $A_r(f)(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|}\frac{1}{n} e^{in\theta}$.

$\Rightarrow A_r(f)(\theta) = \sum_{n=1}^{\infty} \frac{r^n}{n}$ $\quad \lim_{r\to 1} A_r(f)(0) \to +\infty$.

however $|A_r(f)(\theta)| \leq \frac{1}{2\pi}\int_{-\pi}^{\pi} |f(\theta)|\, P_r(\theta) \leq \sup_\theta |f(\theta)|$.

contradiction! ☒

7. Show that $\sum_{n\geq 2} \frac{1}{\log n}\sin nx$ converges for every x. yet itsn't the fourier series.

pf: $\sum_{n\geq 2}^{N} \frac{1}{\log n}\sin nx =$

$\sum_{n=2}^{N} \frac{1}{\log n}\sin nx = \frac{1}{\log N} S_N - \sum_{n=2}^{N-1}\left(\frac{1}{\log(n+1)} - \frac{1}{\log n}\right) S_n$

$S_n = \sum_{k=2}^{n} \sin kx = \frac{1}{2i}\sum_{k=2}^{n}(e^{ikx} - e^{-ikx}) + \frac{1}{2i}\sum_{k=2}^{n}(\omega^k - \omega^{-k})$

$\frac{\sin(n+\frac{1}{2})x}{\sin\frac{x}{2}} < \infty$ $\quad$ when $(n\to\infty)$

So $\sum_{n\geq 2} \frac{1}{\log n}\sin nx$ converges for every x.

Assume. $f\in\mathcal{R}$. such that $f(x) \sim \sum_{n\geq 2} \frac{1}{\log n}\sin nx$.

then $\hat{f}(n) = \frac{\sin nx}{\log n \cdot e^{inx}}$ $\quad (n\geq 2)$

$\sum_{n=2}^{\infty} \frac{r^n}{\log n}\sin nx$ then by Parsevel identity:

$\sum_{n\geq 2}^{\infty} |\hat{f}(n)|^2 < \infty$. that is, $\sum_{n\geq 2}^{\infty} \frac{\sin^2 nx}{(\log n)^2} = \sum_{n\geq 2}^{\infty} \frac{1}{2(\log n)^2} - \sum_{n\geq 2}^{\infty} \frac{\cos 2nx}{(\log n)^2}$

$\to \infty$. ($\log n < \sqrt{n}$ when n is big

$\Rightarrow \frac{1}{(\log n)^2} > \frac{1}{n}$) ☒

8. (a). let f be $f(\theta)=|\theta| \quad \theta\in[-\pi,\pi]$

Use Parseval's identity to caculate: $\sum_{n=0}^{\infty}\frac{1}{(2n+1)^4}$ $\sum_{n=1}^{\infty}\frac{1}{n^4}$

pf: $a_n=\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{\pi}|\theta|e^{-in\theta}d\theta = \begin{cases}\frac{\pi}{2} & \text{if } n=0\\ \frac{-1+(-1)^n}{n^2\pi} & \text{if } n\neq 0\end{cases}$

Since $\|f\|^2=\sum_{n=-\infty}^{\infty}|a_n|^2$

that is: $2\sum_{n=0}^{\infty}\frac{1}{(2n+1)^4\pi^2}+\frac{\pi^2}{4}=\frac{1}{2\pi}\int_{-\pi}^{\pi}\theta^2 d\theta=\frac{\pi^2}{3}$

$\Rightarrow \sum_{n=0}^{\infty}\frac{1}{(2n+1)^4}=\frac{\pi^4}{96}$

$\sum_{n=1}^{\infty}\frac{1}{n^4}=\sum_{n=1}^{\infty}\frac{1}{16n^4}+\sum_{n=0}^{\infty}\frac{1}{(2n+1)^4} \Rightarrow \sum_{n=1}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{90}$ ☒

(b). $f(\theta)=\theta(\pi-\theta)$. Similarly : we have $\sum_{n=0}^{\infty}\frac{1}{(2n+1)^6}=\frac{\pi^6}{960}$ & $\sum_{n=0}^{\infty}\frac{1}{n^6}=\frac{\pi^6}{945}$.

9. Show that for α not an integer, the Fourier series of $\frac{\pi}{\sin\pi\alpha}e^{i(\pi-x)\alpha}$ on $[0,2\pi]$. is given by $\sum_{n=-\infty}^{\infty}\frac{e^{inx}}{n+\alpha}$. Apply Parseval's formula to show that $\sum_{n=-\infty}^{\infty}\frac{1}{(n+\alpha)^2}=\frac{\pi^2}{(\sin\pi\alpha)^2}$.

pf: $\hat{f}(n)=\frac{1}{2\pi}\int_{-\pi}^{2\pi}\frac{\pi}{\sin\pi\alpha}e^{i(\pi-x)\alpha}e^{-inx}dx$

$=\frac{1}{2}\int_{-\pi}^{2\pi}\frac{e^{i\pi\alpha}}{\sin\pi\alpha}e^{-ix(\alpha+n)}dx = \frac{e^{i\pi\alpha}}{2\sin\pi\alpha}\frac{e^{-2\pi i(\alpha+n)}-1}{-i(\alpha+n)}$

$=\frac{1}{n+\alpha}$.

$\Rightarrow f(x)\sim\sum_{n=-\infty}^{\infty}\frac{e^{inx}}{n+\alpha}$

then $\sum_{n=-\infty}^{\infty}\frac{1}{(n+\alpha)^2}=\frac{1}{2\pi}\int_0^{2\pi}\left(\frac{\pi}{\sin\pi\alpha}\right)^2 d\theta=\frac{\pi^2}{(\sin\pi\alpha)^2}$ ☒

11. (a). if f is T-periodic. continuous and piecewise C^1 with $\int_0^T f(t)dt=0$.

Show that $\int_0^T|f(t)|^2dt\le\frac{T^2}{4\pi^2}\int_0^T|f'(t)|^2dt$. "=" $\Leftrightarrow f(t)=A\sin\left(\frac{2\pi t}{T}\right)+B\cos\left(\frac{2\pi t}{T}\right)$

pf: ~~$\int_0^T|f(t)|^2dt=\sum_{n=-\infty}^{\infty}|\hat{f}(n)|^2$~~ ~~$\left(\hat{f}(0)=\int_0^T f(t)dt=0\right)$~~

$=\sum_{n=-\infty}^{\infty}\frac{1}{T^2}\left|\int_0^T f(t)e^{-\frac{2\pi int}{T}}dt\right|^2$

$=\sum_{n=-\infty}^{\infty}\frac{1}{T^2}\left|\int_0^T f'(t)\frac{e^{-\frac{2\pi int}{T}}}{\frac{2\pi in}{T}}dt\right|^2$.

~~$\frac{1}{4\pi^2}\int_0^T|f'(t)|^2dt=\sum_{n=-\infty}^{\infty}$~~

$=\sum_{n\neq 0}\frac{1}{4\pi^2n^2}\left|\int_0^T f'(t)e^{-\frac{2\pi int}{T}}dt\right|^2$,

$\frac{1}{T^2}\int_0^T|f(t)|^2dt=\sum_{n=-\infty}^{\infty}|\hat{f}(n)|^2$. $\hat{f}(0)=0$.

$\frac{1}{T^2}\int_0^T|f'(t)|^2dt=\sum_{n=-\infty}^{\infty}|\hat{f}'(n)|^2=\sum_{n\neq 0}\frac{4\pi^2n^2}{T^2}|\hat{f}(n)|^2\ge\frac{4\pi^2}{T^2}\cdot\frac{1}{T^2}\int_0^T|f(t)|^2dt$

$\Rightarrow\int_0^T|f'(t)|^2dt\ge\frac{4\pi^2}{T^2}\int_0^T|f(t)|^2dt$

When "=" ($n\ge 2$ $\hat{f}(n)=0\Rightarrow f(x)=A\sin\left(\frac{2\pi t}{T}\right)+B\cos\left(\frac{2\pi t}{T}\right)$ ☒

(b). f is above and g is C^1 and T-periodic prove that

$$\left|\int_0^T \overline{f(t)}\, g(t)\,dt\right|^2 \le \frac{T^2}{4\pi^2}\int_0^T |f(t)|^2\,dt \int_0^T |g'(t)|^2\,dt$$

Pf: $\Leftrightarrow |(f,g)|^2 \le \frac{T^2}{4\pi^2}\|f\|^2\,\|g'\|^2$

$(f,g) = \sum_{n=-\infty}^{\infty} a_n \overline{b_n}$ $\quad a_n = \hat{f}(n)$. $\quad b_n = \hat{g}(n)$ $\quad \hat{g'}(n) = \frac{2\pi i n}{T}\hat{g}(n)$

$\Rightarrow |(f,g)|^2 = \left|\sum_{n=-\infty}^{\infty} a_n \overline{b_n}\right|^2$.

$\|f\|^2 = \sum_{n=-\infty}^{\infty} |a_n|^2$. $\quad \|g'\|^2 = \sum_{n=-\infty}^{\infty} \frac{4\pi^2 n^2}{T^2}|b_n|^2 \ge \frac{4\pi^2}{T^2}\sum_{n=-\infty}^{\infty}|b_n|^2$.

$\Rightarrow |(f,g)|^2 \le \frac{T^2}{4\pi^2}\|f\|^2\cdot\|g\|^2$ ☑

(c). $[a,b]\subseteq\mathbb{R}$ $\quad f\in C^1[a,b]$. $\quad f(a)=f(b)=0$

$$\int_a^b |f(t)|^2\,dt \le \frac{(b-a)^2}{\pi^2}\int_a^b |f'(t)|^2\,dt.$$



Pf: expand f to $[2a-b, b]$.

then f is $2(b-a)$-periodic, continuous & $f\in C^1[2a-b,b]$ $\quad \int_{2a-b}^{b} f(t)\,dt = 0$

by (a) we have $\int_a^b |f(t)|^2\,dt = \frac{1}{2}\int_{2a-b}^{b}|f(t)|^2\,dt \le \frac{(b-a)^2}{4\pi^2}\int_{2a-b}^{b}|f'(t)|^2\,dt$

$= \frac{(b-a)^2}{\pi^2}\int_a^b |f'(t)|^2\,dt$ ☑

p.s. $\frac{(b-a)^2}{\pi^2}$ cannot be improved. by (a) we have "=" holds iff $f(t) = A\sin\left(\pi\frac{t-a}{b-a}\right)$ ☑

12. Prove that $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$

Pf: $D_N(\theta) = \sum_{k=-N}^{N} e^{ik\theta} = \frac{\sin(N+\frac{1}{2})\theta}{\sin\frac{\theta}{2}} = 1 + 2\sum_{n=1}^{N}\cos(N\theta)$

$\Rightarrow \int_{-\pi}^{\pi} D_N(\theta)\,d\theta = 2\pi$ $\quad$ let $f(\theta) = \frac{1}{\sin\frac{\theta}{2}} - \frac{2}{\theta}$.

Then $\int_{-\pi}^{\pi} D_N(\theta)\,d\theta = \int_{-\pi}^{\pi}\frac{2\sin(N+\frac{1}{2})\theta}{\theta}\,d\theta + \int_{-\pi}^{\pi}\sin(N+\frac{1}{2})\theta\, f(\theta)\,d\theta$

let $N\to\infty$ by Riemann-Lesgue Lemma: $\int_{-\pi}^{\pi}\sin(N+\frac{1}{2})\theta\, f(\theta)\,d\theta \to 0$ ☑

Then $2\pi = \lim_{N\to\infty}\int_{-(N+\frac{1}{2})\pi}^{(N+\frac{1}{2})\pi}\frac{2\sin\theta}{\theta}\,d\theta \Rightarrow \int_{-\infty}^{+\infty}\frac{\sin\theta}{\theta}\,d\theta = \pi$

$\Rightarrow \int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$ ☑

13 Suppose f is periodic and of class C^k. Show that

$\hat{f}(n) = o\left(\frac{1}{|n|^k}\right)_{(n\to\infty)}$ (Hint: R-L Lemma).

pf $\hat{f}(n) = \frac{1}{T}\int_0^T f(x)\, e^{\frac{-2\pi i n x}{T}}\, dx$.

To make it convenient to write, just take $T = 2\pi$

$$\hat{f}(n) = \frac{1}{2\pi}\int_0^{2\pi} f(x)\, e^{-inx}\, dx = \frac{1}{2\pi in}\int_0^{2\pi} f'(x) e^{-inx}\, dx$$
$$= \dots = \frac{(-1)^{k-1}}{2\pi (in)^k}\int_0^{2\pi} f^{(k)}(x)\, e^{-inx}\, dx$$

$f^{(k)}(x) \in C[0, 2\pi]$ So by Riemann-Lebesgue lemma. $\int_0^{2\pi} f^{(k)}(x) e^{-inx}\, dx \to 0 \quad (n\to\infty)$

$\Rightarrow \hat{f}(n) = o\left(\frac{1}{|n|^k}\right)$ ☐

14. Prove the Fourier series of a continuously differentiable function f on the circle is absolute convergent.

pf: take $T = 2\pi$.

$$\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\, dx \qquad f(x) \sim \sum_{n\neq 0} \frac{1}{2\pi in}\hat{f'}(n) e^{inx} + \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, dx$$
$$= \frac{1}{2\pi in}\int_{-\pi}^{\pi} f'(x) e^{-inx}\, dx$$

then $\left(\sum_{n\neq 0} \frac{1}{|n|}|\hat{f'}(n)|\right)^2 \le \sum_{n\neq 0}\frac{1}{n^2}\sum_{n\neq 0}|\hat{f'}(n)|^2 = \left(\sum_{n\neq 0}\frac{1}{n^2}\right)\left(\|f'\|^2 - \frac{1}{4\pi^2}\left|\int_{-\pi}^{\pi} f'(x)\, dx\right|^2\right)$

$< \infty$.

So $\sum_{n\neq 0}\frac{1}{2\pi in}\hat{f'}(n) e^{inx} + \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)'dx$ is absolutely convergent ☐

15. Let f be 2π-periodic. and, Riemann integrable on $[-\pi, \pi]$.

(a) Show that $\hat{f}(n) = \frac{-1}{2\pi}\int_{-\pi}^{\pi} f(x+\frac{\pi}{n}) e^{-inx}\, dx$.

hence $\hat{f}(n) = \frac{1}{4\pi}\int_{-\pi}^{\pi}[f(x) - f(x+\frac{\pi}{n})] e^{-inx}\, dx$.

pf: $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\, dx \overset{t = x-\frac{\pi}{n}}{=\!=\!=} \frac{1}{2\pi}\int_{-\pi-\frac{\pi}{n}}^{\pi-\frac{\pi}{n}} f(t+\frac{\pi}{n})\, e^{-in\star} e^{-i\pi}\, dt$

$$= \frac{-1}{2\pi}\int_{-\pi-\frac{\pi}{n}}^{\pi-\frac{\pi}{n}} f(t+\tfrac{\pi}{n})\, e^{-int}\, dt$$
$$= \frac{-1}{2\pi}\int_{-\pi}^{\pi} f(t+\tfrac{\pi}{n}) e^{-int}\, dt$$ ☐

(b). Assume f satisfies a Hölder condition of order α

namely: $|f(x+h) - f(x)| \le C\cdot|h|^\alpha$. for some $0 < \alpha \le 1$. some $C > 0$, and all x, h.

Use a) Show that $\hat{f}(n) = O\left(\frac{1}{|n|^\alpha}\right)$

pf: $|\hat{f}(n)| \le \frac{1}{4\pi}\int_{-\pi}^{\pi} \left|f(x) - f(x+\frac{\pi}{n})\right| |e^{-inx}|\, dx$

$$\le \frac{\pi^\alpha}{4\pi}\int_{-\pi}^{\pi} \frac{1}{|n|^\alpha}\, dx = \frac{\pi^\alpha}{2|n|^\alpha} \Rightarrow \hat{f}(n) = O\left(\frac{1}{|n|^\alpha}\right).$$

(c) prove the result cannot be improved by showing that

$f(x) = \sum_{k=0}^{\infty} 2^{-k\alpha} e^{i2^k x}$ where $0<\alpha<1$ satisfies $|f(x+h)-f(x)| < |h|^\alpha$ now

and $\hat{f}(N) = \frac{1}{N^\alpha}$ whenever $N = 2^k$

pf: $|f(x+h)-f(x)| = \left|\sum_{k=0}^{\infty} 2^{-k\alpha}\left(e^{i2^k(x+h)} - e^{i\cdot 2^k x}\right)\right|$

$\leq \underbrace{\sum_{2^k \leq \frac{1}{|h|}} + \sum_{2^k > \frac{1}{|h|}}} \left|2^{-k\alpha}\left(e^{i2^k h} - 1\right)\right|$

estimate two sum by different inequality:

$\leq \sum_{2^k \leq \frac{1}{|h|}} 2^{-k\alpha}\cdot 2^k |h| + \sum_{2^k > \frac{1}{|h|}} \cdot 2^{1-k\alpha}$

$|e^{i\theta}-1| \leq |\theta|$ when $|\theta|<1$ $\leq |h| \frac{2^{(1-\alpha)M}-1}{2^{1-\alpha}-1} + \frac{2^{1-(M+1)\alpha}}{1-2^{-\alpha}}$ where $2^M \leq \frac{1}{|h|}$, $2^{M+1} > \frac{1}{|h|}$

or $|e^{i\theta}-1| \leq 2$ $< C|h|^\alpha$ ... (1).

let $N = 2^m$

$\hat{f}(N) = \frac{1}{2\pi}\int_{-\pi}^{\pi}\left(\sum_{k=0}^{\infty} 2^{-k\alpha}\cdot e^{i2^k x}\right) e^{-i\cdot 2^m\cdot x}\,dx$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_{k=0}^{\infty} 2^{-k\alpha} e^{ix(2^k-2^m)}\,dx$

$= \frac{1}{2\pi}\sum_{k=0}^{\infty} 2^{-k\alpha}\int_{-\pi}^{\pi} e^{ix(2^k-2^m)}\,dx$ $\qquad \int_{-\pi}^{\pi} e^{ixt}\,dx = \begin{cases} 2\pi & t=0 \\ 0 & t\in 2\mathbb{Z}\setminus\{0\} \end{cases}$

$= \frac{1}{2^{m\alpha}} = \frac{1}{N^\alpha}$ ∎

16. Let f be 2π-periodic which satisfies Lipschitz condition with constant K.

(a) For every positive h : define $g_h(x) = f(x+h) - f(x-h)$.

Prove that $\frac{1}{2\pi}\int_0^{2\pi} |g_h(x)|^2\,dx = \sum_{n=-\infty}^{\infty} 4|\sin nh|^2 |\hat{f}(n)|^2$.

and show that $\sum_{n=-\infty}^{\infty} |\sin nh|^2 |\hat{f}(n)|^2 \leq K^2h^2$.

pf: $\hat{g}_h(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(x+h)-f(x-h))\,e^{-inx}\,dx$

$= \frac{1}{2\pi}\left(\int_{-\pi+h}^{\pi+h} f(t)\,e^{-in(t-h)}\,dt - \int_{-\pi-h}^{+\pi-h} f(t)\,e^{-in(t+h)}\,dt\right)$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\left(e^{-in(t-h)} - e^{-in(t+h)}\right)dt$

$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(t)\,e^{-int}\cdot \sin nh\cdot 2i\,dt$

$\Rightarrow |\hat{g}_h(n)| = 2|\sin nh|\,|\hat{f}(n)|$

Parseval $\Rightarrow \frac{1}{2\pi}\int_0^{2\pi} |g_h(x)|^2\,dx = \sum_{n=-\infty}^{\infty} 4|\sin nh|^2 |\hat{f}(n)|^2$.

$\frac{1}{2\pi}\int_0^{2\pi} |f(x+h)-f(x-h)|^2\,dx \leq 4K^2h^2$ ∎

(b) let p be a positive integer. By choosing $h = \frac{\pi}{2^{p+1}}$

show that $\sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)|^2 \le \frac{K^2\pi^2}{2^{2p+1}}$

pf. $\frac{1}{2}\sum_{2^{p-1}<|n|\le 2^p} |f(n)|^2 \le \sum_{2^{p-1}<|n|\le 2^p} |\sin nh|^2 |\hat{f}(n)|^2 < K^2h^2 = \frac{K^2\pi^2}{2^{2p+2}}$ $\left(\sin nh > \frac{\sqrt{2}}{2}\right)$

(c) Estimate $\sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)|$, then conclude that the fourier series of f converges absolutely thus. $S_N(f) \to f$ uniformly

pf: $\left(\sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)|\right)^2 \overset{\text{Cauchy}}{\le} 2^p \sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)|^2 < \frac{K^2\pi^2}{2^{p+1}}$

thus $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| = \lim_{p\to\infty} \sum_{2^{p-1}<|n|\le 2^p} |\hat{f}(n)| + |\hat{f}(0)|$

$< K\pi \left(\sum_{p=0}^{\infty} 2^{-\frac{p+1}{2}} \right) + |\hat{f}(0)| < \infty$ ∎

(d). If f satisfies a Hölder condition of order $\alpha > \frac{1}{2}$, then the fourier series converges absolutely (proof is similar).

17. If f is a bounded monotonic function on $[-\pi, \pi]$ then $\hat{f}(n) = O(\frac{1}{n})$.

pf: $\hat{f}(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx$ Assume $f \uparrow$ and $|f| \le M$.

Lemma: $g(x) = \chi_{[a,b]}(x)$ $\hat{g}(0) = \frac{b-a}{2\pi}$ $\hat{g}(n) = \frac{1}{2\pi i n}\left(e^{-ina} - e^{-inb}\right)$

thus. $\hat{g}(n) = O(\frac{1}{|n|})$

let $h(x) = \sum_{k=1}^{N} \alpha_k \chi_{[a_k, a_{k+1}]}(x)$ with $-\pi = a_1 < \dots < a_{N+1} = \pi$. & $-M \le \alpha_1 \le \dots \le \alpha_N \le M$

we have $|\hat{h}(n)| = \left|\sum_{k=1}^{N} \alpha_k \frac{1}{2\pi i n}\left(e^{-ina_k} - e^{-ina_{k+1}}\right)\right|$ using Abel means:

$\le \frac{M}{2\pi|n|}\left|\sum_{k=1}^{N}\left(e^{-ina_k} - e^{-ina_{k+1}}\right)\right|$ $\chi \overset{\text{Abel means}}{=\!=} \frac{1}{2\pi n}\left|\sum_{k=1}^{N-1}(\alpha_{k+1}-\alpha_k)e^{-ina_{k+1}} + \alpha_N\left(1 - e^{-ina_N}\right)\right|$

$= \frac{M}{2\pi n}$. So $\hat{h}(n) = O(\frac{1}{|n|})$ uniformly in N $\le \frac{C}{2\pi n}$

Back to the proof: given $N > 0$. N is big enough

fix $-\pi = a_1 < \dots < a_{N+1} = \pi$, $\alpha_k = \sup f[a_k, a_{k+1}]$ $\beta_k = \inf f[a_k, a_{k+1}] = f(a_k)$

note that $\|T\| = \sup_k |a_{k+1} - a_k| = f(a_{k+1})$

let $h_N(x) = \sum_{k=1}^{N} \alpha_k \chi_{[a_k, a_{k+1}]}(x)$ $g_N(x) = \sum_{k=1}^{N} \beta_k \chi_{[a_k, a_{k+1}]}(x)$

then $h_N(x) \ge f(x) \ge g_N(x)$

$|\hat{f}(n)| = |\widehat{f - g_N}(n) + \hat{g}_N(n)| \le |\widehat{f - g_N}(n)| + |\hat{g}_N(n)|$

let $N \to \infty$ and $\|T\| \to 0$ we have $(f - g_N)(x) \to 0$ uniformly in x

thus $\widehat{f - g_N}(n) \to 0$ uniformly in n $\Rightarrow |\hat{f}(n)| \le \frac{C}{2\pi n} \Rightarrow \hat{f}(n) = O(\frac{1}{n})$

18. We have proved

(a) if f is of class C^k then $\hat{f}(n) = o\left(\frac{1}{|n|^k}\right)$

(b) if f is Lipschitz then $\hat{f}(n) = O\left(\frac{1}{|n|}\right)$ & $\sum |\hat{f}(n)|$ converges absolutely

(c) if f is monotonic then $\hat{f}(n) = O\left(\frac{1}{|n|}\right)$

(d) if f satisfies Hölder condition with order of α $(0<\alpha<1)$
then $\hat{f}(n) = O\left(\frac{1}{|n|^\alpha}\right)$

(e) if f is integrable (Riemann), then $\sum |\hat{f}(n)|^2 < \infty$ thus $\hat{f}(n) = o(1)$.

prove. For every $\{\varepsilon_n\}_n \to 0$ $\varepsilon_n \geq 0$, there exists a continuous function f such that $|\hat{f}(n)| \geq \varepsilon_n$ for infinitely many values of n.

pf: choose a sebsequence of $\{\varepsilon_n\}_n$
such that $\sum_k \varepsilon_{n_k} < \infty$

then Choose $f(x) = \sum_{k=-\infty}^{\infty} a_k e^{\pm i n_k x}$ where $a_k = \varepsilon_{n_k} + \frac{1}{2^k}$

then $\hat{f}(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \sum_{k=-\infty}^{\infty} a_k e^{\pm i n_k x} \cdot e^{-imx}\, dx$

$= \pm \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} a_k \int_{-\pi}^{\pi} e^{(i n_k x \mp imx)}\, dx$

$= \begin{cases} a_k & m = n_k \quad k = 1, 2, \dots \\ 0 & m \neq n_k \end{cases}$

(p.s. $\sum_{n=-\infty}^{\infty} |\hat{f}(n)| = \sum_{k=1}^{\infty} |a_k| < \infty$, So f is continuous)

So $|\hat{f}(n_k)| > \varepsilon_{n_k}$ ▨

19. Give another proof that the sum $\sum_{0<|n|\leq N} \frac{e^{inx}}{n}$ is uniformly bounded (but don't have a limit) in N and x.

pf: $\frac{1}{2i} \sum_{0<|n|\leq N} \frac{e^{inx}}{n} = \sum_{n=1}^{N} \frac{\sin nx}{n} = \frac{1}{2} \int_0^x (D_N(t) - 1)\, dt$

(p.s. $D_N(t) = \sum_{n=-N}^{N} e^{inx} = 1 + 2\sum_{n=1}^{N} \cos nx \Rightarrow \int_0^x (D_N(t) - 1) = 2 \sum_{n=1}^{N} \frac{\sin nx}{n}$.)

then $\left| \int_0^x (D_N(t) - 1)\, dt \right| = \left| \int_0^x \left(\frac{\sin(N+\frac{1}{2})t}{\sin \frac{t}{2}} - 1 \right) dt \right| < \pi|x| + \left| \int_0^x \sin D_N(t)\, dt \right|$

$< \pi|x| + \left| \int_0^x \frac{\sin(N+\frac{1}{2})t}{t}\, dt \right| + \int_0^x \left(\frac{1}{t} - \frac{1}{\sin \frac{t}{2}} \right) \sin(N+\frac{1}{2})$

$\leq \pi + \int_0^{\infty} \frac{\sin x}{x} dx + C < \infty$ ▨

20. $f(x)=\frac{\pi-x}{2}$ $x\in(0,2\pi)$. with $f(0)=0$ extend it to $\mathbb{R}$

The fourier series of f is $f(x)\sim\sum_{|n|\neq 0}\frac{e^{inx}}{2in}=\sum_{n=1}^{\infty}\frac{\sin nx}{n}$

Show that $\lim_{N\to\infty}\left(\max_{0<x\le\frac{\pi}{N}} S_N(f)(x)-\frac{\pi}{2}\right)=\int_0^{\pi}\frac{\sin t}{t}dt-\frac{\pi}{2}$

$$S_N(f)(x)=\frac{1}{2}\sum_{0<|n|\le N}\frac{e^{inx}}{n}=\frac{1}{2}\int_0^x(D_N(t)-1)dt. \qquad 0<x\le\frac{\pi}{N}$$

where $D_N(t)=\frac{\sin(N+\frac{1}{2})t}{\sin\frac{t}{2}}$ $\Rightarrow \frac{dS_N(f)(x)}{dx}=\frac{1}{2}(D_N(x)-1)$

$$=\frac{1}{2\sin\frac{x}{2}}\left(\sin(N+\tfrac{1}{2})x-\sin\tfrac{x}{2}\right)=0$$

$\Rightarrow x=\frac{\pi}{N+1}<\frac{\pi}{N}$ (c)

then $\max S_N(f)(x)=\frac{1}{2}\int_0^{\frac{\pi}{N+1}}(D_N(t)-1)dt$

$$=\frac{1}{2}\int_0^{\frac{\pi}{N+1}}\left(\frac{\sin(N+\frac{1}{2})t}{\sin\frac{t}{2}}-1\right)dt=\sum_{n=1}^{N}\frac{1}{n}\sin\left(\frac{n\pi}{N+1}\right)$$

now we caculate $\lim_{N\to\infty}\sum_{n=1}^{N}\frac{1}{n}\sin\left(\frac{n\pi}{N+1}\right)=\lim_{n\to\infty}\sum_{n=1}^{N}\frac{\sin\left(\frac{n\pi}{N+1}\right)}{\frac{n\pi}{N+1}}\frac{\pi}{N+1}$

$$=\int_0^{\pi}\frac{\sin x}{x}dx.$$

So $\lim_{N\to\infty}\left(\max_{0<x\le\frac{\pi}{N}} S_N(f)(x)\right)=\int_0^{\pi}\frac{\sin x}{x}dx$ ∎